



WORKSHOP PROJECT

PROJECT REPORT



Project Title:

Design and Fabrication of a Multi-Functional Interchangeable Head Hammer

Course:	Workshop Practice / Manufacturing Lab
Department:	Mechanical Engineering

GROUP MEMBERS

NAME:	Tayyab Mahmood Adil	NAME:	Muhammad Usman Basit
ROLL NO.	2502272	ROLL NO.	2502287
CLASS:	BEME-F-25-B	CLASS:	BEME-F-25-B

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SUBMITTED TO:
SIR RAHEEL SHERAZ

ABSTRACT

In daily life, hammers are commonly used tools for striking objects, inserting nails, and performing basic mechanical tasks. Conventional hammers are usually limited to one or two operations and are not suitable for different surface hardness levels. This project focuses on the design and fabrication of a multi-functional hammer with interchangeable heads. The hammer is manufactured using PTFE (Teflon) and mild steel, employing various workshop operations such as facing, turning, threading, drilling, taper turning, and chamfering. The unique feature of this hammer is its internally threaded head, which allows different attachments to be installed for use on hard and soft surfaces. The project successfully demonstrates the practical application of workshop machines and tools while solving a real-life problem through an innovative design.

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INTRODUCTION

Hammers are among the most commonly used hand tools in daily life as well as in workshops. They are used for striking, shaping, inserting nails, and assembling components. However, most conventional hammers are designed for specific tasks and are not suitable for use on different materials such as wood, metal, or marble. Using the wrong hammer can damage the workpiece or reduce efficiency.

This project aims to design a multi-functional hammer that can be used for multiple applications by changing its striking head. The hammer uses an internally threaded head system that allows attachments of different materials to be fixed according to the surface requirement. This design increases tool versatility and reduces the need for multiple hammers.

LITERATURE REVIEW / BACKGROUND

Hammers are among the most widely used hand tools in workshops, construction work, and daily life applications. Different types of hammers are designed for specific tasks. For example, claw hammers are commonly used for driving and removing nails in wood, while ball peen hammers are used in metalworking operations such as shaping and riveting. Rubber mallets and soft-face hammers are preferred when working on delicate surfaces to avoid surface damage.

Although these hammers are effective for their individual purposes, each type has limited functionality. A hard hammer can damage soft or brittle materials such as marble and wood, whereas a soft hammer cannot be used for heavy striking operations on metal surfaces. Due to these limitations, users are often required to use multiple hammers for different tasks, which increases tool cost, storage requirements, and inconvenience. Existing hammer designs available in the market generally focus on single-purpose applications and do not provide a simple mechanism for adapting the tool to different surface hardness levels. This creates a need for a more versatile and practical solution.

The present project addresses this limitation by designing and fabricating a multi-functional hammer with interchangeable heads. By using a threaded mechanism, different attachments can be installed based on the application. This approach improves versatility, reduces the need for multiple tools, and provides an efficient solution for workshop and daily-life use.

PROBLEM STATEMENT

The conventional hammer used in daily life can perform only limited operations, typically striking nails or objects of similar hardness. It is not suitable for use on different surfaces such as soft wood, metal, or brittle materials like marble. This limitation creates the need for multiple hammers, increasing cost and inconvenience. Therefore, there is a need to design a single hammer that can perform multiple tasks efficiently by using interchangeable striking heads.

WORK DISTRIBUTION AMONG GROUP MEMBERS

Tayyab Mahmood Adil:

- Concept development and idea generation
- Lathe machine operations on PTFE (facing, turning, taper turning)
- Threading and assembly work
- Final finishing and report preparation

Muhammad Usman Basit:

- Material cutting and preparation
- Drilling operations using radial drill machine
- Metal part fabrication and grinding
- Assistance in assembly and testing



METHODOLOGY

The project was carried out by first identifying a real-life problem related to limited hammer functionality. A design solution using interchangeable threaded heads was proposed. Material selection was done based on weight, strength, and machinability. PTFE was selected for the handle and hammer head, while metal was used for hard surface attachments. Machining operations were then performed using lathe machines, drilling machines, and grinding tools. Finally, all components were assembled and finished to obtain the final prototype.

MATERIALS USED (BILL OF QUANTITIES)

Sr. No	Material	Specification	Quantity
1.	PTFE (Teflon) Rod	Diameter 35mm, Length 279.4mm	1
2.	Metal Rod	Diameter 37mm, Length 70.8mm	1
3.	Sand Paper	Fine and Medium	As Required
4.	Oil	Cutting Oil	As Required
5.	Wood (For Base)	310mm x 265mm, thickness 17mm	1
6.	Metal Sheet	Length 160mm, Thickness 2mm	1

THEORETICAL MODELLING (CALCULATIONS + AUTOCAD MODEL)

- Total length of PTFE rod = 11 inches = 279.4 mm
- Handle length after facing = 160 mm
- Hammer head length after facing = 100 mm
- Handle diameter = 35 mm
- External thread on handle = M17.5mm
- Internal thread on hammer head = M17.5mm
- Internal threads on both faces of head = M16mm
- AutoCAD software was used to prepare 3D and 2D drawings of the hammer assembly, handle, head, and metal attachments with accurate dimensions.

FABRICATION PROCEDURE

1. The PTFE rod was cut into two pieces of lengths 165 mm (handle) and 105 mm (head).
2. These pieces were faced on a small lathe to achieve a smooth surface.
3. After the facing the length of the handle is 160mm and length of head is 100mm.
4. Handle diameter and Head diameter remains same i.e 35mm.
5. After this, the handle diameter was reduced to 18mm to the length of 35mm.
6. External threading of M17.5 was performed on the handle using a tap wrench.
7. A central hole was drilled (drill bit 16mm) in the hammer head using a radial drill machine.
8. Internal threading of M17.5 was performed inside the hammer head.
9. Taper turning was carried out on the handle at an angle of 20.5° for a length of 20 mm.
10. Hammer head was drilled from both faces (25.4mm deep)
11. And internal threading of M16 was performed.
12. Steel rod (diameter 37mm) was cut into two small pieces (length 38mm).
13. These pieces were faced (after facing length was 35.4mm), turned (2mm).
14. After this, these pieces were again turned (up to length 25.4mm) to reduce diameter to 17mm.
15. Then, these pieces were threaded (M16 using tap wrench), chamfered, and finished.
16. All parts were assembled and surface finishing was done using files and sand paper.

MODEL DEVELOPMENT (REAL-TIME IMAGES)

Handle



Cutting



Facing



Turning



Threading

Head



Facing



Drilling



Internal Threading

Faces



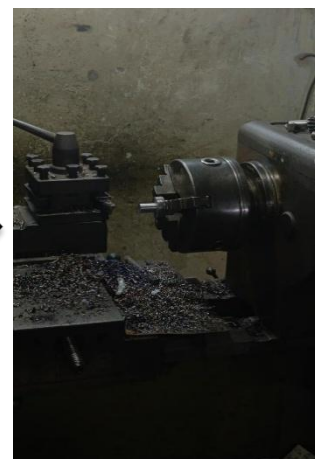
Cutting



Smoothing



Facing



Turning / Threading

PROTOTYPE EXPERIMENT RESULTS (FUNCTIONAL, OBSERVATIONS AND RESULTS)

The fabricated hammer was tested on wood, metal, and marble surfaces. The steel attachments performed well on hard surfaces, while the PTFE head reduced surface damage on softer materials. The threaded mechanism allowed easy attachment and removal of different heads. The hammer functioned successfully as a multi-purpose tool.

DISCUSSION

The project successfully achieved its objective of creating a multi-functional hammer. The use of PTFE reduced weight and vibration, while steel attachments provided strength for hard applications. Although some challenges were faced due to machine and tool limitations, proper planning and teamwork helped in completing the project effectively.

RECOMMENDATIONS

- Additional head attachments of rubber or nylon can be added.
- Knurling can be performed on the handle for better grip.
- Heat-treated steel can be used for improved durability.
- Precision CNC machining can improve accuracy and finish.

CONCLUSION

The design and fabrication of a multi-functional hammer was successfully completed. The project solved a practical daily-life problem by introducing interchangeable hammer heads suitable for different materials. This project enhanced practical knowledge of machining operations, teamwork, and problem-solving skills. The final prototype worked effectively and met all design objectives.

REFERENCES

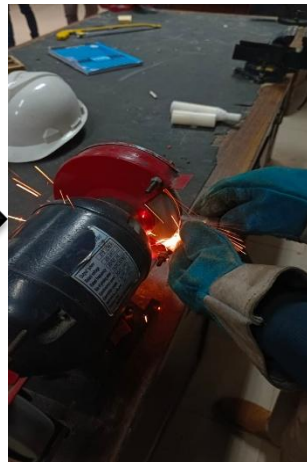
[Literature Review](#)

APPENDICES

BASE



Cutting



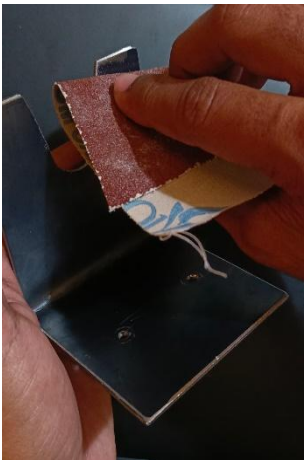
Smoothing



Drilling



Bending



Filing / Finishing



Wood



Final Product

PROCEDURE

1. Cut the metal sheet of thickness 2mm up to length 170mm.
2. After cutting sheet may melted from sides, making it sharp. Use pedestal grinder to smooth the surfaces.
3. After smoothing the length will be 160mm.
4. Place this metal sheet on drill machine and drill two holes of diameter 4mm.
5. Bend this sheet from center on bending machine at 90 degree.
6. Use scriber to draw V-shape and cut it using hacksaw.
7. Use file and sand paper for finishing.
8. Take a wood of length 310mm, width 265mm and of thickness 17mm.
9. Draw a line from its center using pen.
10. Draw the lines on 45 degrees (to make it hexagonal).
11. Cut the extra part (using hacksaw) and use sand paper for finishing.
12. Take a drill bit of M4 and drill two hole on wooden base.
13. Fix metal stand on base using

SOLIDWORKS 3D DRAWING

3D DRAWING

